

Final Assessment- 2025, Fall

Level: Bachelor

Programme: BE(Computer)

Course: Dat Science and Analytics

Year/Part: IV/I

Set-B

Full Marks : 100

Pass Marks : 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

1. (a) Explain data preprocessing techniques:

- i. Data cleaning
- ii. Data integration
- iii. Data transformation
- iv. Discretization [8]

(b) A numeric variable has the following values:

12.4, 25.6, 31.2, 45.8, 28.9, 39.5, 41.3, 22.7, 18.6, 35.4

- i. Standardize the data.
- ii. Construct a histogram and comment on distribution shape. [8]

OR

Why Use a Data Analysis Pipeline? Differentiate between Structured and unstructured data in appropriate examples.

2. (a) Define numeric and categorical variables and give 3 examples of each from real life datasets. Describe the differences between normal, exponential, and power law distributions with examples. [8]

OR

Cross-tabulation example: 20 people surveyed for Gender (M/F) and Preference (Tea/Coffee): [8]

	Tea	Coffee
Male	5	5
Female	3	7

a) Create the cross-tabulation table.

v) Compute the chi-squared value to test independence.

(b) The following data represents age (X) and annual income (Y in thousands):

X	22	25	30	35	40
Y	25	30	45	50	60

- i. Compute Pearson correlation coefficient.
- ii. Interpret the result.
- iii. Test significance using t-test. [7]

3. a) Explain R^2 and Adjusted R^2 in model evaluation. Discuss multicollinearity and Variance Inflation Factor (VIF). [8]

b) You are given the following data:

x	0.5	1.0	1.5	2.0
y	1.2	2.3	3.1	3.8

Estimate the value of y at x = 1.2 using the Nadaraya-Watson estimator. Take bandwidth $h = 0.3$ and Gaussian kernel. [7]

OR

Given a dataset with n variables, explain how Principal Component Analysis (PCA) is computed. How is the variance explained by each component determined? [7]

4.a) Consider a simple linear regression model: [8]

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Given:

X	1	2	3	4	5
Y	3	5	7	9	11

i) Write the matrix form of the regression model.

ii) Estimate β_0 and β_1 . iii) Predict Y when $X = 6$.

b) Describe the CART (Classification and Regression Trees) algorithm in detail. Explain how the Gini Index is used as a splitting criterion and how ~~CART~~ handles categorical variables. [7]

5. (a) Explain permutation-based variable importance and partial dependence plots. [8]

(b) Perform one step of K-means clustering for the following data with $k=2$
Points: (1,2), (2,1), (8,9), (9,8) [7]

Initial centroids: (1,2) and (8,9).

6. (a) What is time series analysis? Explain the concepts of autocorrelation and stationarity, and discuss why stationarity is important in time series modeling. [7]

(b) What is Granger causality? What is a Causal Directed Acyclic Graph (DAG)? Explain its main components and how DAGs help in representing causal relationships. [8]

OR

Explain the importance of selecting an optimal lag length in time series models. Discuss how Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are used for this purpose. [8]

7. Write short notes on (Any Two): [2×5= 10]

a) Statistical significance (p-value, t-value) b) Logistic Regression

c) Causation vs Correlation

*****All The Best*****